

Member 4 Task Sheet

QA, Accessibility Developer and Presentation Lead

Author: Project Lead | Date: July 26, 2026 | IEEE Hackathon

1. Core Role Focus

You own quality assurance, accessibility, and presentation materials. Your role is to write automated test files, program browser-based Text-to-Speech audio readouts, mock WhatsApp message reminders, compile slide decks, and test runs.

2. Directories to Create and Manage

- **backend/data/** Store mock test documents.
- **backend/tests/** Write endpoint verification code.
- **frontend/src/components/** Create TTS audio players.

3. Target File Checklists & Instructions

A. **backend/data/sample_summons.pdf & eviction_notice.png (Day 1 - Highest Priority)**

Create realistic notice templates containing complex dates and legal sections to test the parser's validation limits.

B. **backend/tests/test_api.py (Day 2 - High Priority)**

Develop automated python test suites using pytest. Check upload routing parameters, database operations, and RAG vector searches.

C. **backend/services/sms_scheduler_mock.py (Day 3/4 - Low Priority)**

Build mock scheduler outputting terminal alerts to simulate automated SMS/WhatsApp alerts near deadlines.

D. frontend/src/components/VoiceReader.tsx (Day 4 - High Priority)

Create a React voice widget using browser Web Speech API. Include Play, Pause, Stop buttons, volume slider, and reading speed controls.

E. README.md & presentation_guide.md (Day 4 - High Priority)

Write installation guides, organize pitch slide timelines, and capture screen backup demo videos.

4. Daily Schedule Checklist

- **Day 1:** Design mock PDF/image notices to serve as test case inputs.
- **Day 2:** Write automated pytest verification scripts validating endpoints.
- **Day 3:** Create WhatsApp reminder mockup slide charts.
- **Day 4:** Build Web Speech TTS audio reader widgets and compile slides.